

BEARING CAPACITY OF SHALLOW FOUNDATION

A Parametric Study Using Excel

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Introduction:

The bearing capacity of soil is an important consideration in the design of shallow foundations. A foundation must safely transfer structural loads to the underlying soil without excessive settlement or shear failure. In this project, the ultimate and safe bearing capacities of shallow foundations are evaluated for different soil conditions using standard bearing-capacity equations. Excel is used to perform the calculations and present the variation of bearing capacity graphically.

Objective:

- >To understand the concept of bearing capacity of shallow foundations.
- > To calculate the bearing-capacity factors N_c , N_q , and N_γ .
- > To calculate ultimate bearing capacity for different soil conditions.
- > To calculate safe bearing capacity using a factor of safety.
- > To study the effect of foundation width on bearing capacity using Excel.

Theory

Add the main equation you used:

$$q_u = cN_c + \gamma D_f N_q + \frac{1}{2} \gamma B N_\gamma$$

Where:

- q_u = ultimate bearing capacity
- c = cohesion
- ϕ = angle of internal friction
- γ = unit weight of soil
- D_f = depth of foundation
- B = foundation width
- N_c, N_q, N_γ = bearing-capacity factors

For $\phi > 0$:

$$N_q = e^{\pi \tan \phi} \tan^2 \left(45^\circ + \frac{\phi}{2} \right)$$
$$N_c = \frac{N_q - 1}{\tan \phi}$$

And the expression we used for N_γ :

$$N_\gamma = 2(N_q + 1) \tan \phi$$

For the $\phi = 0^\circ$ clay case, we used:

$N_c=5.7, N_q=1, N_\gamma=0$

Excel-Based Calculation:

Bearing Capacity Analysis of Shallow Foundation										
Soil	Cohesion, c (kPa)	Friction Angle, ϕ (°)	Unit Weight, γ (kN/m ³)	Foundati on Width, B (m)	Foundati on Depth, Df (m)	Nc	Nq	Ny	Ultimate Bearing Capacity, q_u (kPa)	Safe Bearing Capacity, q_{safe} (kPa)
Loose Sand	0	30	16	2	1	30.1396278	18.40112222	22.40248627	652.8577358	217.6192453
Medium Dense Sand	0	35	17	2	1	46.1235987	33.29609149	48.02876359	1382.522536	460.8408454
Soft Clay	20	0	17	2	1	5.7	1	0	131	43.66666667
Medium Clay	30	0	18	2	1	5.7	1	0	189	63

Foundati on Width B (m)	Loose Sand	Medium Dense Sand	Soft Clay	Medium Clay
1	157.8793	324.7593	43.66667	63
1.5	187.7493	392.8001	43.66667	63
2	217.6192	460.8408	43.66667	63
2.5	247.4892	528.8816	43.66667	63
3	277.3592	596.9223	43.66667	63

->The calculations were performed in Microsoft Excel. The input soil parameters were entered into the spreadsheet and the bearing-capacity factors, ultimate bearing capacity, and safe bearing capacity were calculated using Excel formulas.

Variation of Bearing Capacity with Foundation Width:

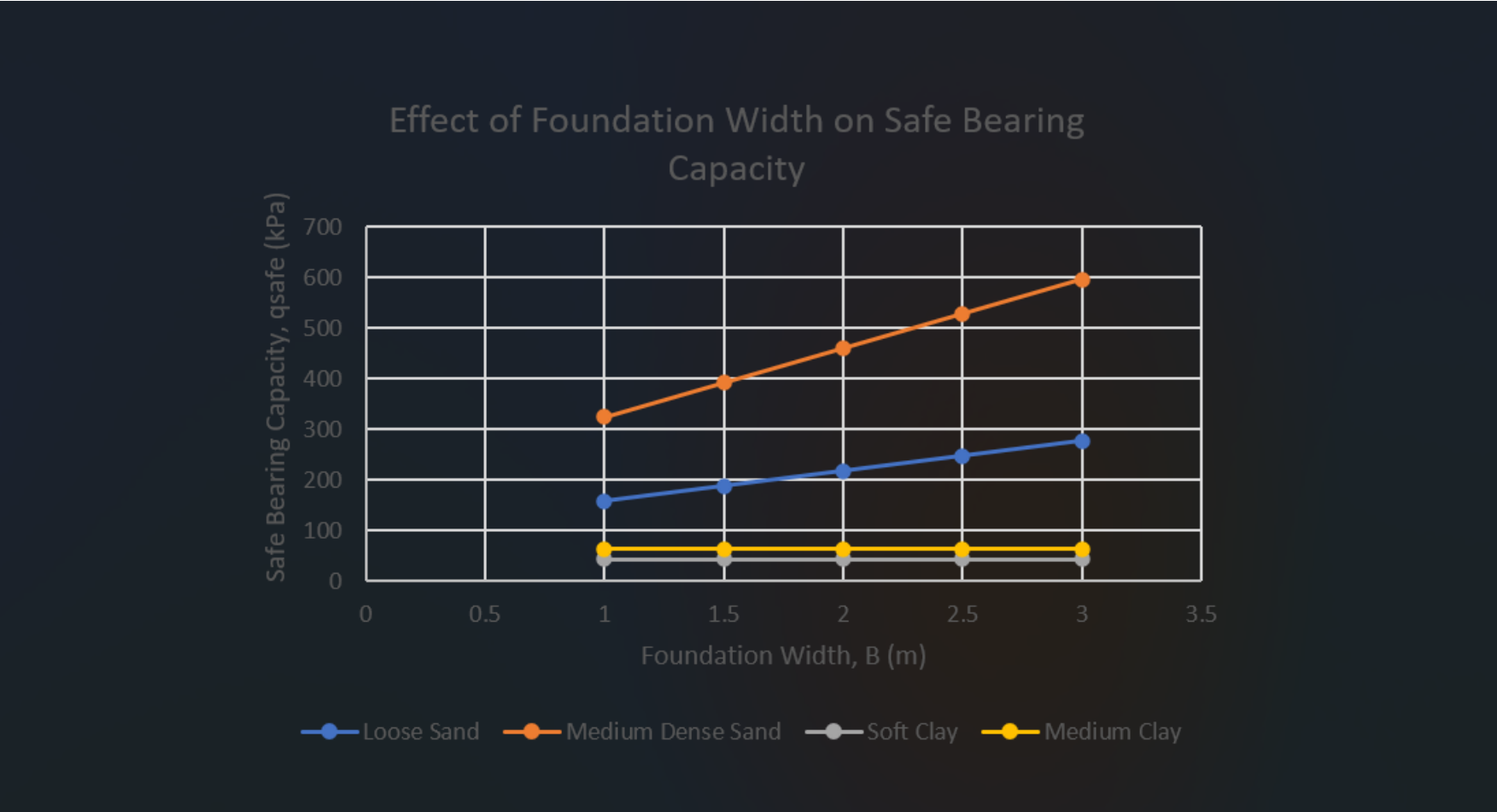


Figure 1: Variation of Ultimate Bearing Capacity with Foundation Width

Soil Type	Ultimate Bearing Capacity q_u	Safe Bearing Capacity q_{safe}
Loose Sand	652.86	217.62
Medium Dense Sand	1382.52	460.84
Soft Clay	131.00	43.67
Medium Clay	189.00	63

Conclusion:

The bearing capacity of shallow foundations was evaluated for four different soil conditions using Excel-based calculations. The results show a clear variation in bearing capacity with soil type.

Among the four cases, **Medium Dense Sand** exhibited the highest ultimate bearing capacity of **1382.52 kN/m²** and a safe bearing capacity of **460.84 kN/m²**. **Loose Sand** showed an ultimate bearing capacity of **652.86 kN/m²** and a safe bearing capacity of **217.62 kN/m²**. For the clayey soils, **Medium Clay** had an ultimate bearing capacity of **189.00 kN/m²** and a safe bearing capacity of **63.00 kN/m²**, while **Soft Clay** showed the lowest values, with an ultimate bearing capacity of **131.00 kN/m²** and a safe bearing capacity of **43.67 kN/m²**.

The results indicate that the soil condition and shear-strength parameters have a significant influence on the bearing capacity of a shallow foundation. The comparison shows that the sand cases considered in this study provide higher bearing capacities than the clay cases, with Medium Dense Sand giving the most favorable result. The graphical representation further helps visualize the variation in bearing capacity for the different foundation conditions.

Overall, the project demonstrates how **Excel can be effectively used to perform and present shallow-foundation bearing-capacity calculations**, while providing a simple basis for comparing different soil conditions.